

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2– Concept Mapping

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COURSE NAME: Data Structures

COURSE CODE: CSA0306

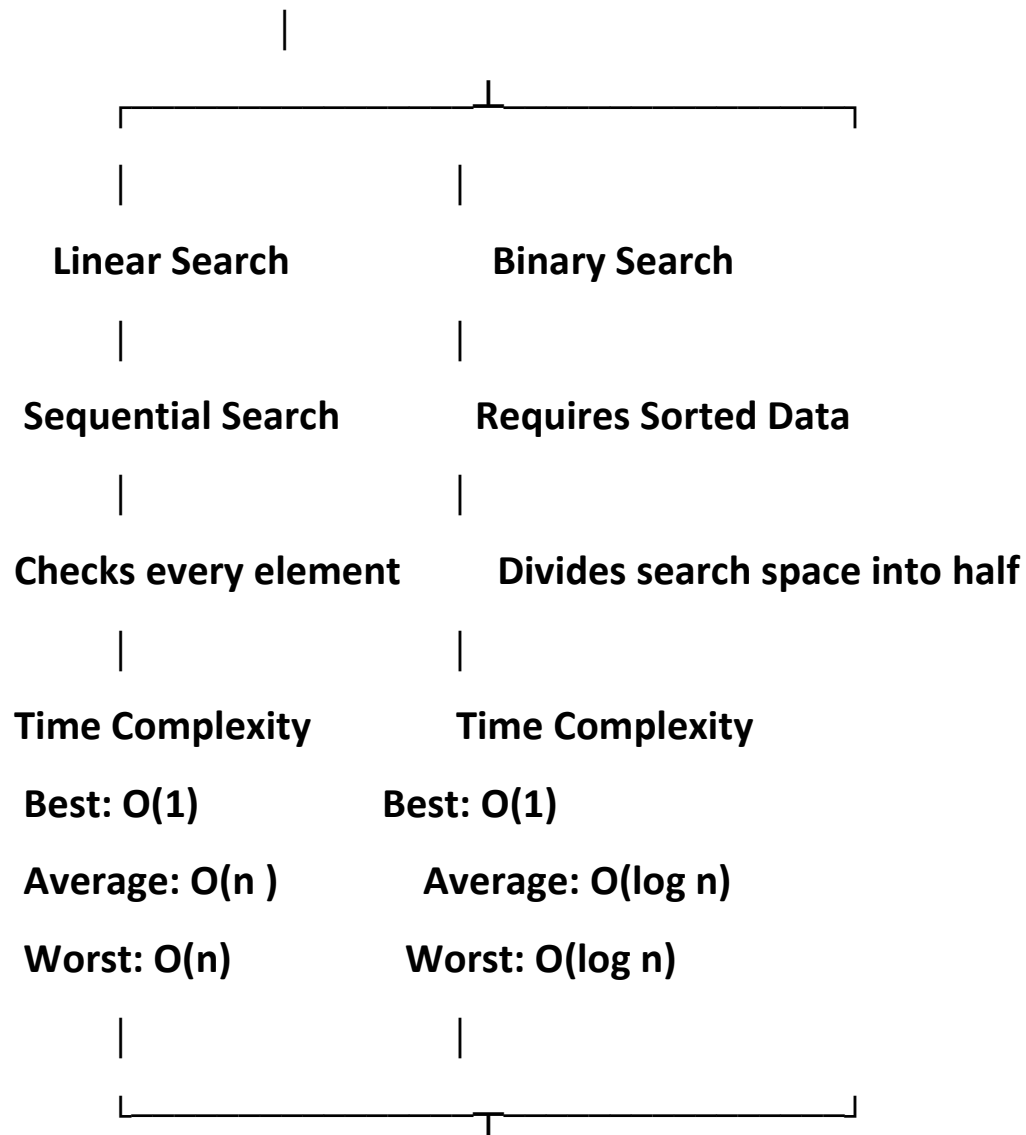
Code of Conduct

I Lochana Sri R.S. (192521348) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Lochana Sri R.S.

1. A system needs to search for specific records in a large dataset efficiently. Construct a concept map linking Searching Algorithm → Linear Search → Binary Search → Time Complexity → Efficiency. Include relationships that compare the number of operations and performance of each method. Analyze which algorithm is more efficient and justify your conclusion.

SEARCHING ALGORITHM



Efficiency

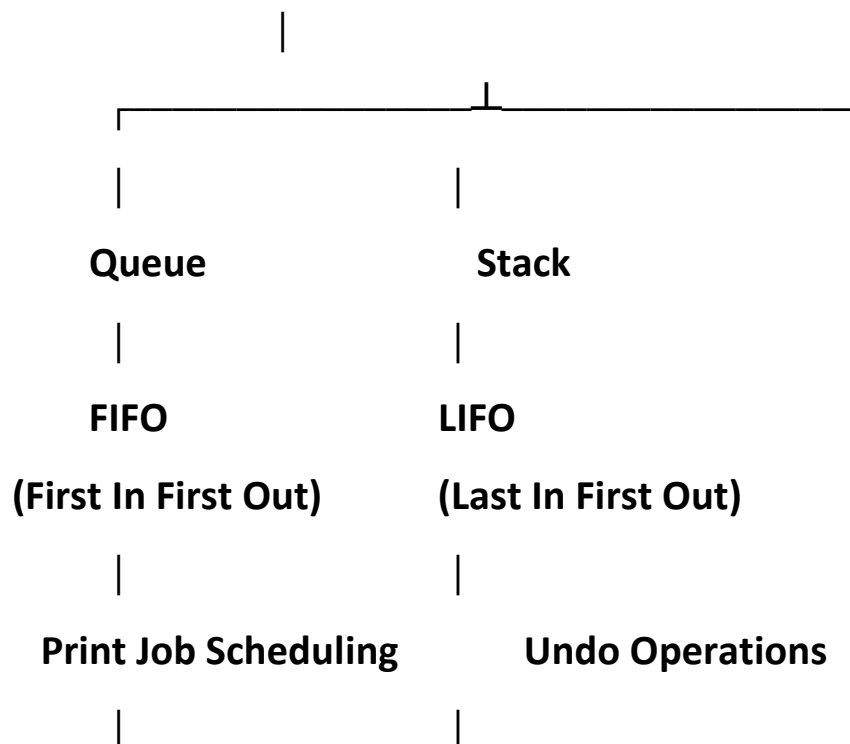
Binary Search is more efficient for large sorted datasets because it performs fewer comparisons.

Explanation:

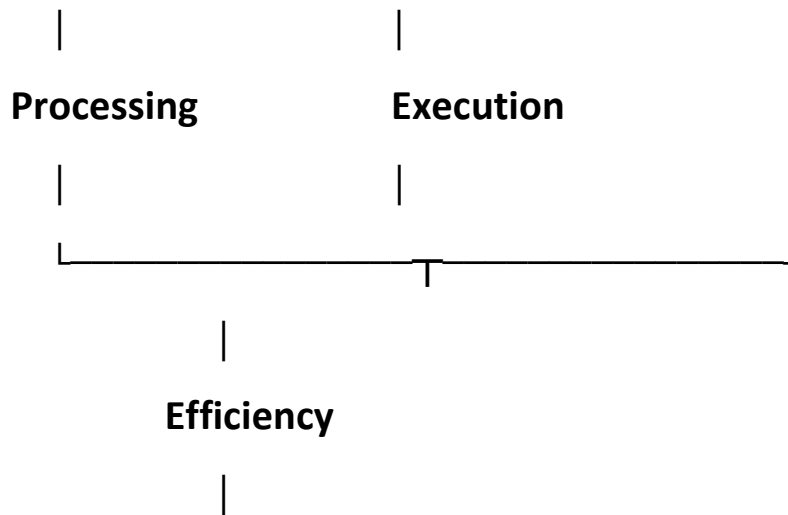
Searching algorithms are used to find a particular element in a dataset. Linear Search checks each element one by one until it finds the required item, so it works even if the data is not sorted. Binary Search works only on sorted data and finds the element by repeatedly dividing the dataset into two halves. The efficiency of these algorithms is measured using time complexity. Linear Search has a worst-case time complexity of $O(n)$, while Binary Search has $O(\log n)$. Therefore, Binary Search is faster and more efficient for large sorted datasets, whereas Linear Search is better for small or unsorted data.

2. A printer system manages print jobs submitted by multiple users. Construct a concept map connecting Queue $\rightarrow$ FIFO $\rightarrow$ Job Scheduling $\rightarrow$ Processing $\rightarrow$ Efficiency and Stack $\rightarrow$ LIFO $\rightarrow$ Undo Operations $\rightarrow$ Execution. Explain how each structure is applied and justify their suitability in this context.

DATA STRUCTURES



Jobs processed in order Last action removed first



**Queue ensures fair job scheduling
while Stack provides quick undo
functionality.**

Explanation:

A Queue follows the FIFO (First In, First Out) principle, where the first print job submitted is processed first. This makes it suitable for printer job scheduling because all jobs are handled in the correct order. A Stack follows the LIFO (Last In, First Out) principle, where the most recent operation is performed first. This is useful for Undo operations because the last action can be reversed easily. Both data structures are efficient for their specific purposes. Queue is the best choice for managing print jobs, while Stack is ideal for implementing undo functionality.